

DATA MANAGEMENT CARPENTRY · EPISODE 7

What to Store and When to Remove

Retention mandates, digital waste, reproducible archiving, and safe deletion practices across desktop and server platforms.

EPISODE SCOPE

Questions & Objectives

Core Questions

Which data should I keep vs. delete?

How long am I required to retain data?

What are the hidden costs of hoarding?

How does deletion work on servers vs. desktop?

Learning Objectives

Identify key retention factors & mandates.

Distinguish raw inputs from redundant scratch.

Perform permanent deletion safely across systems.

Recognise digital waste and overcome hoarding fear.

THE CORE DILEMMA

Too Much Data, Too Little Clarity

The Hoarder's Rut

> *"We are running out of space, and we don't know what half of these files actually are."*

It is easy to adopt a *"keep everything just in case"* mindset. We generate temporary versions, duplicate exports, and test plots until a digital swamp forms.

The Hidden Repercussions

- Search fatigue finding authoritative files
- Running pipeline scripts on stale drafts
- Security liability keeping expired personal data
- Exceeding institutional drive quotas

DISCUSSION 1 · 5 MINUTES

To Keep or To Delete?

Imagine you are cleaning out a research project directory from two years ago:

Working in pairs or small groups: What factors or criteria would you consider when deciding whether to **retain** a file or **permanently delete** it?

- Can the data easily be regenerated?
- What was the financial acquisition cost?
- Does it contain sensitive personal data?

DECISION FRAMEWORK

The Retention Checklist: 4 Factors

1. Funder & Institutional Mandates

UKRI, NIH, and universities mandate retaining primary data for **5 to 10 years** post-publication.

2. Reproducibility & Provenance

Irreplaceable raw physical runs MUST be kept. Intermediate tables can be pruned if scripts exist.

3. Legal & Privacy Constraints

GDPR mandates deleting personal data once consent expires. Subjects have statutory right to erasure.

4. Data Integrity & Documentation

Files without headers, units, dates, or metadata are scientific noise and pose severe error risks.

EXERCISE · 5 MINUTES

Challenge: Evaluating the Messy Dataset

Decide whether each file should be RETAINED or DELETED:

1. raw_sequencing_data_2024_03_12.fastq (Original machine output from a £5,000 lab run)
2. temporary_merged_subset_v2.csv (Temporary file used to test a quick plotting idea)
3. analysis_script_final.py (Python script used to clean raw data into publication figures)
4. draft_paper_comments_John.docx (Old manuscript draft from last year with tracked changes)
5. unlabelled_test_results.txt (Text file of numbers, no headers, no date, unknown author)

CHALLENGE SOLUTION

Audit Decisions & Justifications

RETAIN	`raw_sequencing_data.fastq`	Irreplaceable, costly raw data. Mandated by funders for post-pub reproducibility.
DELETE	`temporary_merged_subset_v2.csv`	Temporary scratchpad. Easily recreated if pipeline code and raw files are kept.
RETAIN	`analysis_script_final.py`	Essential methodology recipe. Tiny disk footprint but maximum scientific value.
ARCHIVE	`draft_paper_comments_John.docx`	Obsolete once published. If kept for records, isolate into an admin archive/ folder.
DELETE	`unlabelled_test_results.txt`	Zero headers or provenance. Dangerous noise that risks polluting valid future runs.

OS MECHANICS

Local Deletion: The Recycle Bin Safety Net

Clicking Delete is a Move Operation

In Windows File Explorer or macOS Finder, pressing delete simply moves files into hidden staging folders (Recycle Bin or .Trash).

Files can easily be restored to their original location if removed by mistake.

The Space Reclaim Trap

The file still occupies disk space! If your hard drive is full, pressing delete frees zero capacity until the bin is emptied.

Action: Right-click bin icon > Select Empty Recycle Bin / Empty Trash.

SECURITY HYGIENE

Secure Data Scrubbing & Residual Remanence

Magnetic Hard Drives (HDDs)

Standard deletion removes index pointers; raw sector data remains until overwritten by new files.

Overwriting tools: shred (Linux), cipher /w (Windows), or terminal tools overwrite sector bits directly.

⚠

Solid State Drives (SSDs) & Disposal

SSDs use wear-leveling controllers, rendering traditional block-shredders unreliable.

Best protection: Whole-disk encryption + certified institutional confidential waste disposal.

ENTERPRISE STORAGE

Remote Server Deletion: No Safety Net

Immediate & Permanent

On institutional file stores (RDW) or Linux compute clusters, **there is no Recycle Bin**.

Files and pointers are erased instantly upon confirmation; disk space is reclaimed immediately without an undo buffer.

Operational Safeguards

- **Double-check targets:** Verify working directory path before purging.
- **Manual staging:** Move files into a to_delete/ staging folder first.
- **Snapshot awareness:** Confirm backup frequency and snapshot restore points.

BEHAVIORAL & ENVIRONMENTAL FACTORS

The Hoarding Trap & Digital Carbon Footprints

The Hoarding Psychology

"What if I delete something important from a £10,000 experiment?"

Fear causes researchers to hoard, leading to mental clutter, search drag, and privacy exposure on unmonitored participant files.

Digital Waste Is Physical

Every gigabyte kept forever spins in a data center requiring 24/7 electrical power and massive cooling systems.

Pruning dead files and duplicate scratchpads directly reduces your research carbon footprint.

DISCUSSION 2

Mitigating Deletion Anxiety

What strategies can you use to safely manage and overcome worries about deleting data?

DISCUSSION 2

Mitigating Deletion Anxiety

What strategies can you use to safely manage and overcome worries about deleting data?

Staged Quarantine

Move files to a dated staging folder; delete after 3 months if unreferenced.

Code-First Assurance

Verify scripts recreate clean outputs deterministically before deleting tables.

Read-Only Raw Data

Lock write permissions on raw files so accidental keystrokes cannot delete them.

SUMMARY

Episode 7 Key Points

Funder Mandates: Preserve primary research records for mandated periods (5–10 years).

High vs. Low Value: Keep raw data and analysis code; systematically purge scratch files.

Local vs. Server: Desktop bins must be emptied to free space; server deletion is permanent.

Hygiene & Sustainability: Routine pruning eliminates mental drag and digital waste.

The Invariant: Over-retaining data creates mental clutter, increases security risks, and contributes to environmental digital waste.

LOOKING AHEAD

Next: Episode 8

We now know how to prune, organize, and retain files safely. Next, we explore tabular data formatting, structuring spreadsheets, and tidy data principles.

Structuring Tabular Data